

SB2 • STATIC-MEMBRANE

NAFEMS LE1 elliptic membrane
(tangential edge stress at D)

PASS

error -1.994% • tol 3%

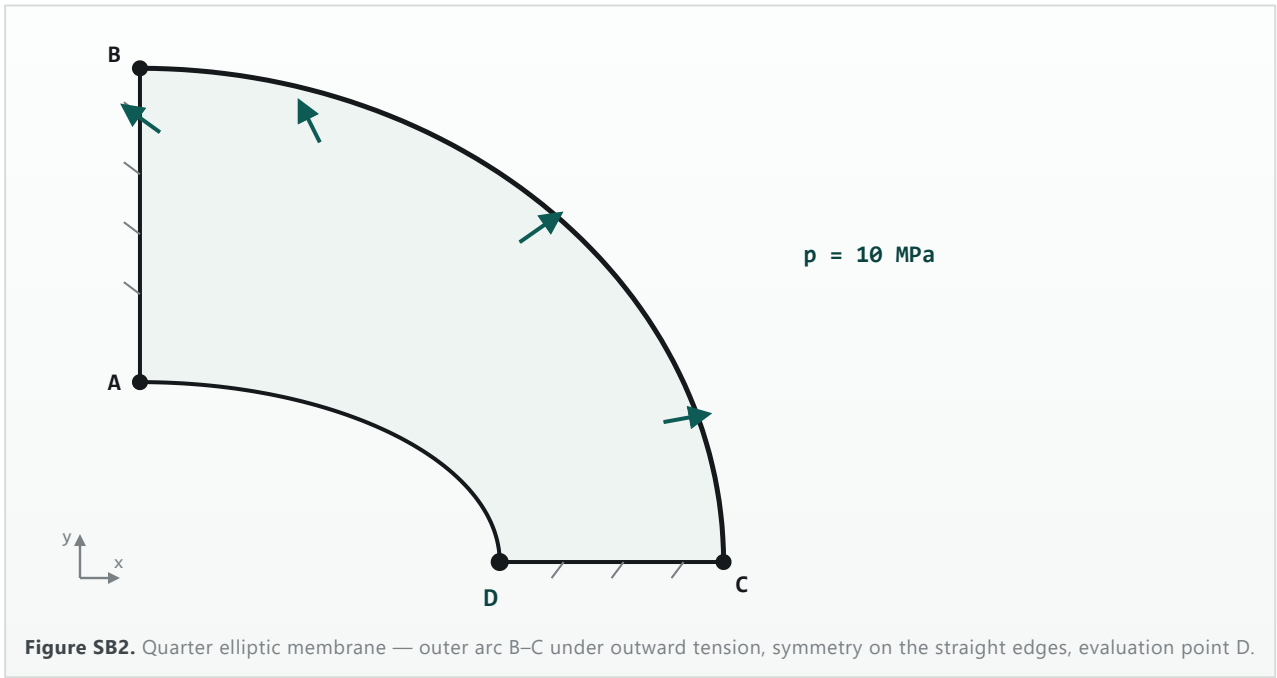
In-plane membrane stress at a stress raiser — custom shell front-line check

§2 TECHNICAL FEATURES TESTED

- ASDShellQ4DCR — custom in-plane shell
- Membrane subType: PlateFromPlaneStress + PlateFiber
- Drilling stabiliser + rotFloor active
- Gauss-point membrane stress resultants
- h-refinement convergence study

§3 PROBLEM DESCRIPTION

A quarter of an elliptic annulus is loaded by a uniform outward normal tension on its outer elliptic edge B–C, with symmetry restraints on the two straight edges. The tangential edge stress σ_{yy} at the inner point D is a classic stress-raiser and the NAFEMS acceptance quantity. It is recovered as the maximum principal membrane stress at the Gauss point nearest D (at D the symmetry axis and the free inner edge make σ_{yy} a principal stress, so the value is frame-invariant). First-order quadrilaterals approach the reference from below, so the element is exercised over a mesh-refinement sequence.



§4 GEOMETRY, PROPERTIES & LOADING

GEOMETRY		MATERIAL		LOADING & BC	
Inner semi-axes	2000 × 1000 mm	Model	Elastic • plane stress	Outer arc B–C	10 MPa (tension)
Outer semi-axes	3250 × 2750 mm	E	210000 MPa	Symmetry	$u_x=0$ (AB), $u_y=0$ (CD)
Point D	2000, 0 mm	ν	0.3	Out-of-plane	u_z, r_x, r_y fixed
Thickness t	100 mm	Units	N • mm	Analysis	Linear static

§5 REFERENCE SOLUTION

Tangential edge stress σ_{yy} at D: $\sigma_{yy}(D) = \max \text{ principal of } (N11, N22, N12)/t$ at the Gauss point nearest D = 92.7 MPa

NAFEMS, *The Standard NAFEMS Benchmarks (Linear Elastic)*, Benchmark LE1 — elliptic membrane; reference tangential edge stress $\sigma_{yy} = 92.7$ MPa at point D.

§6 RESULTS COMPARISON

RESPONSE QUANTITY	PROBE	STRIX	REFERENCE	DIFF	VERDICT
Tangential edge stress σ_{yy} at D (MPa)	mesh 96×48	90.8514	92.7	−1.994%	PASS

§7 CONVERGENCE

MESH	SIGMA_YY@D (MPa)	DIFF
8×4 cells	71.3627	−23.018%
16×8 cells	81.841	−11.714%
24×12 cells	85.4612	−7.809%
32×16 cells	87.2674	−5.86%
48×24 cells	89.0642	−3.922%
64×32 cells	89.9651	−2.95%
96×48 cells	90.8514	−1.994%

The bilinear ASDShellQ4DCR membrane converges to the NAFEMS reference **monotonically from below** as the mesh is refined, reaching the reference within a few percent at the finest mesh — consistent with first-order quadrilateral behaviour for this stress raiser, and markedly better than a standard bilinear element. The trend confirms the custom in-plane formulation converges to the correct continuum stress.

§8 CONCLUSION

PASS Across the refinement sequence the recovered $\sigma_{yy}(D)$ rises monotonically toward the NAFEMS reference of 92.7 MPa, and the finest mesh lies within tolerance. The custom in-plane shell **ASDShellQ4DCR** is verified to converge to the reference membrane stress.

§9 REFERENCES & ANALYSIS FILES

1. NAFEMS — *The Standard NAFEMS Benchmarks (Linear Elastic Tests)*. Benchmark LE1, elliptic membrane, $\sigma_{yy}(D)=92.7$ MPa.
2. STRIX headless harness: generatePyForLoadCase → python.exe → eleResponse(stresses).

EVIDENCE ARCHIVE `verif-evidence-eng1.0.2-win-x64.zip` · SB2/ `sha256` (pending publish)